

TOI-2046b

R-2243

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI2046_230828 · created 2026-08-15T23:15:22+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460184.78 +/- 0.13 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.099 +/- 0.04
Transit depth (Rp/Rs)^2	0.97 +/- 0.78 [%]
Orbital Inclination (inc)	83.27 +/- 2.74
Airmass coefficient 1 (a1)	1.031 +/- 0.019
Airmass coefficient 2 (a2)	-0.019 +/- 0.019
Scatter in the residuals of the lightcurve fit is	1.82 %
Best Comparison Star	#2 - [315, 280]
Optimal Aperture	4.5
Optimal Annulus	13.15
Transit Duration (day)	0.04 +/- 0.038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.099 ± 0.04 vs 0.1213 (deviation 0.46σ)
Duration fit vs published	0.04 d vs 0.1033 d (deviation 1.37σ)
Mid-time O–C	0.5 min
Depth SNR	2.0
Residual scatter	1.82 %

Light curve

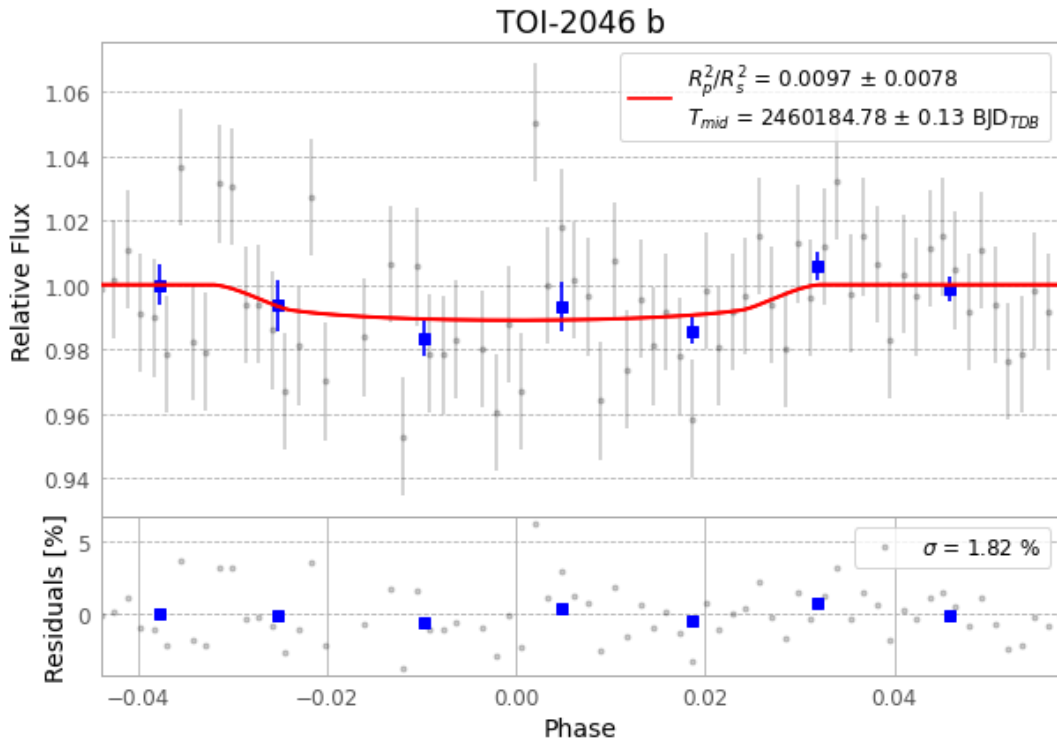


Figure 1 — FinalLightCurve_TOI-2046 b_2023-08-27.png (SHA-256 44abfffd3aaab706...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [299, 248], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ee15c6cb540b5963...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

74 FITS frames (49 MB), TOI2046230828050315.FITS → TOI2046230828084215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI2046-230828.log (2a05fc0c3d3d...), FinalLightCurve_TOI-2046 b_2023-08-27.png (44abfffd3aaa...), FinalLightCurve_TOI-2046 b_2023-08-27.csv (2fd0fdfe4e12...)

Compute resources

Wall time 92.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-15 23:15Z.